

Satellite-Based Monitoring of Slow Ground Deformation in the Chimanimani District, Zimbabwe: An InSAR and Machine Learning Approach

Project Report

Group Members

Romeo Thando Dube R252000T

Midlands State University
Department of Mathematics and Computational Sciences
BSc Applied Mathematics and Computational Science

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Abstract

This report documents the design of a satellite radar-based ground deformation monitoring project for the Chimanimani District, Zimbabwe, an area with a documented history of rainfall-triggered landslides and debris flows, most notably during Cyclone Idai in March 2019. The project applies the Small Baseline Subset (SBAS) InSAR technique to Sentinel-1 satellite imagery to generate a displacement time series for the study area, combined with a statistical anomaly detection layer to flag deformation behaviour that departs from established baseline patterns. This report consolidates the problem definition, the methods used by comparable studies internationally and regionally, the mathematical basis of the chosen approach, the decisions reached by the group to date, and an honest account of the method's capabilities and limitations, including its unsuitability for real-time emergency response without supplementary ground-based instrumentation.

1 Introduction

Regions of Zimbabwe with steep or mountainous terrain, particularly the Eastern Highlands, host infrastructure and settlements exposed to slow-onset ground deformation as well as sudden, rainfall-triggered slope failure. Existing monitoring in these areas is largely reactive: intervention follows visible damage rather than anticipating it. This project applies Interferometric Synthetic Aperture Radar (InSAR) time-series analysis, an established remote sensing technique capable of detecting surface displacement at millimetre-scale precision, to a defined study area within Zimbabwe, in order to build a monitoring and anomaly detection pipeline grounded in an actual documented hazard history rather than a hypothetical one.

2 Problem Statement

There is no continuous, quantitative ground deformation monitoring system currently operating for Zimbabwe's landslide-prone terrain. This gap became starkly visible during Cyclone Idai in March 2019, when debris flows and landslides in the Chimanimani District killed more than 340 people nationally, displaced over 51,000, and destroyed approximately 1,500 km of roads. The project addresses this gap by developing a reproducible InSAR-based pipeline capable of quantifying surface displacement over time and flagging deformation anomalies that may indicate elevated slope failure risk, using Chimanimani as the primary study area given its documented hazard profile and available before/after satellite reference imagery.

3 Documented Precedent: Why Chimanimani

The choice of study area is grounded in a specific, verifiable event rather than a general assumption of risk.

- **Cyclone Idai, March 2019.** Satellite imagery (Landsat 8, Operational Land Imager) captured before and after the cyclone shows extensive landslide scarring across the Chimanimani District, with numerous small-to-moderate landslide events combining to overwhelm river systems and generate destructive debris flows further downstream.
- **Human and infrastructure cost.** Two pupils and a staff member were killed when a landslide struck a secondary school dormitory in the district. Homes in Chimanimani town

were swept away by mudslides, and roads across the wider affected region were destroyed, cutting off communities for extended periods.

- **Terrain factors.** Researchers attributed the severity of the landslide activity partly to the area's steep topographic relief compared to surrounding lowland areas, and to weathered, easily mobilised soils, both consistent with conditions favourable to InSAR-detectable slow deformation between major events.

This gives the project a concrete validation opportunity that a hypothetical or arbitrarily chosen study area would not: known landslide scar locations, an approximate event date, and existing satellite reference imagery that can be used to sanity-check the InSAR pipeline's output.

4 Methods Used by Comparable Studies

The pipeline design draws directly on established methodology rather than being developed from first principles. The following studies were reviewed and directly inform the chosen approach.

4.1 Foundational SBAS Method (Berardino et al., 2002)

The originating SBAS paper introduced a technique for combining multiple small-baseline differential interferograms, linked via singular value decomposition, to recover a continuous surface deformation time series from otherwise disconnected SAR acquisition subsets. This is the mathematical basis of the entire pipeline and is described in full in Section 7.

4.2 SBAS Overview and Refinement (Lanari et al., 2007)

This paper provides a more accessible overview of the SBAS algorithm following its initial development, clarifying implementation details around atmospheric filtering and resolution trade-offs. It is used here as the more approachable secondary reference for the group's own understanding of the method, alongside the original 2002 paper.

4.3 Regional Precedent: Murang'a and Nyeri Counties, Kenya

This study applied SBAS-InSAR to landslide-prone terrain in Kenya, a setting with rainfall-driven slope instability and limited existing monitoring infrastructure broadly comparable to the Zimbabwean context. It demonstrates that the SBAS approach is both technically viable and practically informative when applied to East African terrain, and is the closest regional precedent identified for this project.

4.4 Anomaly Detection Layering: Quetta, Pakistan and Tianjin, China

Both studies combine InSAR-derived deformation fields with a secondary statistical or machine-learning analysis layer to flag hazard-relevant anomalies rather than presenting raw deformation maps alone. The Quetta study pairs SBAS-InSAR output with a machine-learning susceptibility classifier; the Tianjin study applies deformation field analysis specifically in mountainous terrain, closer in character to Chimanimani than flat urban settings. Both inform the design of the anomaly detection layer described in Section 7.3.

4.5 Processing Software: MintPy (Yunjun, Fattahi, and Amelung, 2019)

Rather than implementing phase unwrapping and time-series inversion from scratch, the project uses MintPy, an open-source Python package that implements the SBAS weighted network inversion described mathematically in Section 7. This is standard practice; no reviewed study in this area implements the core InSAR mathematics independently.

5 Aim and Objectives

Aim: To develop and validate a satellite-based ground deformation monitoring and anomaly detection pipeline for the Chimanimani District, Zimbabwe.

Objectives:

- i. Acquire and process a Sentinel-1 SAR image stack for the Chimanimani study area using the SBAS method.
- ii. Generate a displacement time series for coherent surface points within the study area.
- iii. Develop and apply a statistical anomaly detection method to flag deformation inconsistent with established baseline trends.
- iv. Validate flagged anomalies against optical imagery and known landslide scar locations from the 2019 Cyclone Idai event.
- v. Assess the method's applicability, limitations, and appropriate use case (screening and early warning, not real-time emergency response) for Zimbabwean terrain.

6 Group Decisions Reached to Date

The following decisions were reached collectively during project scoping and are recorded here for reference.

Study area: Chimanimani District, selected on the basis of documented landslide history and available reference imagery, rather than an arbitrary or unverified location.

Geothermal angle: Explored and dropped. Initial framing linked the project to geothermal energy potential via "Nyanga geysers"; verification found no geysers in Nyanga and no volcanic mechanism behind Zimbabwe's genuine thermal springs (Rupise, Nyanyadzi, Binga), which are fault-associated rather than tectonically active. This angle was removed from the project scope to avoid an unsupportable claim.

Real-time capability: Explicitly scoped out of the core deliverable. Satellite InSAR is fundamentally retrospective due to orbital revisit constraints (Sentinel-1 revisits the same ground roughly every 6–12 days) and cannot support real-time emergency response. This limitation is stated directly rather than implied or hidden (see Section 8).

Future work framing: A two-stage system is proposed for future extension beyond this project's scope: satellite InSAR for wide-area, low-cost triage of at-risk slopes, followed by targeted deployment of ground-based real-time sensors (Ground-Based InSAR, GNSS stations, or low-cost IoT tiltmeters) at the specific sites flagged as highest-risk. This is presented as a future direction, not a claimed capability of the present pipeline.

7 Mathematical Basis

7.1 Interferometric Phase Model

For two SAR acquisitions at times t_A and t_B , the differential interferometric phase is modelled as

$$\delta\phi = \frac{4\pi}{\lambda} \Delta d_{LOS} + \delta\phi_{\text{topo}} + \delta\phi_{\text{atm}} + \delta\phi_{\text{noise}},$$

where Δd_{LOS} is the line-of-sight surface displacement, λ is the Sentinel-1 wavelength (5.55 cm), and the remaining terms represent residual topographic and atmospheric phase contributions, corrected using a Digital Elevation Model and atmospheric filtering respectively.

7.2 SBAS Time-Series Inversion

Given N SAR acquisitions and M interferometric pairs formed under small spatial and temporal baseline constraints, the unknown mean velocities $\mathbf{v}$ between consecutive acquisitions are recovered from

$$\mathbf{A}\mathbf{v} = \frac{\delta\phi}{4\pi/\lambda},$$

solved in the least-squares sense as

$$\hat{\mathbf{v}} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \left(\frac{\delta\phi}{4\pi/\lambda} \right),$$

with cumulative displacement obtained by temporal integration of $\hat{\mathbf{v}}$.

7.3 Anomaly Flagging

For each coherent pixel, a baseline deformation trend $\hat{d}_i(t)$ is estimated from the displacement time series $d_i(t)$. A point is flagged as anomalous where

$$|d_i(t) - \hat{d}_i(t)| > k\sigma_i,$$

with σ_i the pixel's characteristic deformation variability and k a threshold, typically 2 to 3, determined empirically during validation.

8 Capabilities and Limitations

8.1 What the Pipeline Can Deliver

Pre-event seasonal risk screening (identifying slopes with accelerating deformation ahead of the rainy season); post-event monitoring of previously disturbed slopes, which often remain hazardous for years after an initial failure; and a reusable susceptibility mapping layer for use in seasonal disaster management planning.

8.2 What the Pipeline Cannot Deliver

Real-time or near-real-time emergency response. Sentinel-1's orbital revisit cycle and SBAS processing time make same-day or same-week deformation output impossible in principle, not merely as a matter of insufficient resources. Additionally, coherence loss in areas of dense vegetation is expected to reduce point density across parts of the study area, and residual atmospheric noise remains despite correction.

8.3 Future Extension Toward Active Monitoring

Should the group or a future project wish to pursue real-time capability, this requires a different sensing modality entirely: Ground-Based InSAR, continuous GNSS stations, or IoT-based tiltmeter networks installed at specific high-risk sites identified through the satellite screening stage. This is documented as a future work direction, not a claimed feature of the current project.

9 Timeline

Phase	Duration
Study area scoping and confirmation (Chimanimani)	Week 1
Data acquisition (Sentinel-1 stack, GLO-30 DEM)	Weeks 2–3
InSAR processing (SBAS pipeline via MintPy)	Weeks 4–6
Anomaly detection development	Weeks 7–9
Validation against Idai landslide scars, report writing	Weeks 10–11

10 Team Roles

- **Processing and modelling:** InSAR pipeline execution (MintPy/SBAS) and anomaly detection development.
- **Data and area preparation:** Sentinel-1 and DEM acquisition, file organisation, study area boundary definition.
- **Validation and reporting:** cross-referencing flagged anomalies against optical imagery and known 2019 landslide scar locations; report writing and literature documentation.

11 Expected Outcomes

A validated displacement time series for the Chimanimani study area; a working anomaly detection method flagging deformation of potential concern; a validation exercise against a real, documented landslide event rather than synthetic or assumed ground truth; and a final report documenting methodology, results, and an honest account of the method's limitations and appropriate operational scope.

References

- [1] Berardino, P., Fornaro, G., Lanari, R., and Sansosti, E. (2002). A new algorithm for surface deformation monitoring based on small baseline differential SAR interferograms. *IEEE Transactions on Geoscience and Remote Sensing*, 40(11), 2375–2383. <https://doi.org/10.1109/TGRS.2002.803792>
- [2] Lanari, R., Casu, F., Manzo, M., Zeni, G., Berardino, P., Manunta, M., and Pepe, A. (2007). An overview of the Small BAseline Subset algorithm: a DInSAR technique for surface deformation analysis. *Pure and Applied Geophysics*, 164, 637–661. <https://doi.org/10.1007/s00024-007-0192-9>
- [3] Osmanoglu, B., Sunar, F., Wdowinski, S., and Cabral-Cano, E. (2016). Time series analysis of InSAR data: methods and trends. *ISPRS Journal of Photogrammetry and*

- Remote Sensing*, 115, 90–102. <https://faculty.fiu.edu/~swdowins/publications/Osmanoglu-et-al-ISPRS-2016.pdf>
- [4] SBAS-InSAR analysis for deformation monitoring in landslide-prone areas: A case study of Murang’a and Nyeri Counties in Kenya (2025). *Nova Geodesia*. <https://www.novageodesia.ro/index.php/ng/article/view/423>
- [5] SBAS-InSAR Analysis of tectonic derived ground deformation and subsidence susceptibility mapping via machine learning in Quetta City, Pakistan (2024). *International Journal of Digital Earth*. <https://www.tandfonline.com/doi/full/10.1080/17538947.2024.2441926>
- [6] Three-dimensional surface deformation field monitoring and influencing factors analysis in mountainous areas based on SBAS-INSAR technology, Tianjin, China (2025). *Scientific Reports*. <https://www.nature.com/articles/s41598-025-10894-2>
- [7] Yunjun, Z., Fattahi, H., and Amelung, F. (2019). Small baseline InSAR time series analysis: unwrapping error correction and noise reduction. *Computers & Geosciences*. <https://doi.org/10.1016/j.cageo.2019.104331> (MintPy software: <https://github.com/insarlab/MintPy>)
- [8] Floods and Landslides in Chimanimani (2019). NASA Earth Observatory. <https://science.nasa.gov/earth/earth-observatory/floods-and-landslides-in-chimanimani-144739/>
- [9] Cyclone Idai: Zimbabwe school hit by landslide (2019). BBC News. <https://feeds.bbc.co.uk/news/world-africa-47602399>
- [10] Cyclone Idai triggered serious landslides and debris flows in Zimbabwe (2019). AGU Landslide Blog. <https://blogs.agu.org/landslideblog/2019/03/20/cyclone-idai-1/>